Seventh Implementation Iteration: Space Allocation

with Required Space Features

Software Design / Iteration Report

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1. Iteration Goal

The goal of this seventh implementation iteration is to extend the prototype by adding support for required space features in allocation requests. In previous iterations, the system evaluated requests only according to participant count and time-based availability.

In this iteration, a request may include a required feature such as `Projector`, `Whiteboard`, or `Computers`. A request is approved only if the requested space satisfies the capacity requirement, the required feature requirement, and the availability rule.

2. Changes from Iteration 6

Compared with the sixth iteration, the following changes were introduced:

- Extended the `Space` model by adding supported features:
  - `Projector`
  - `Whiteboard`
  - `Computers`
- Extended one-time and recurring requests by adding a required feature field
- Added a new decision rule: `FeatureRule`
- Updated the request-processing logic so that the requested space must satisfy the required feature before allocation
- Preserved the same request types:
  - `OneTimeRequest`
  - `RecurringRequest`

- Preserved the same supported space types:
  - `Classroom`
  - `Laboratory`
  - `MeetingRoom`
- Preserved the same existing rules:
  - `AvailabilityRule`
  - `CapacityRule`

3. Scenario

A user submits either a one-time request or a recurring request for a specific space resource. Each request may now include a required feature such as a projector, whiteboard, or computers.

The system checks whether the requested space supports the required feature, whether the space capacity is sufficient for the participant count, and whether the requested time slot or all requested occurrences are available.

A one-time request is approved if its single occurrence passes all rules. A recurring request is approved only if all occurrences pass all rules. Otherwise, the request is rejected.

4. Modules and Their Tasks

A. Request Management Module

- Receive one-time or recurring space requests from the user
- Create and store `OneTimeRequest` or `RecurringRequest`
- Include request details:
  - requester
  - requested space
  - requested time slot or time slots with day information
  - participant count
  - required feature
- Send the request to the Rule Evaluation Module

B. Resource Management Module

- Manage all supported space types
- Provide information for `Classroom`, `Laboratory`, and `MeetingRoom`
- Provide identity, type, name, capacity, and supported features of the requested space

C. Rule Evaluation Module

- Apply `CapacityRule`
- Apply `FeatureRule`
- Apply `AvailabilityRule`
- For recurring requests, evaluate all requested occurrences
- Return the final decision: Approved or Rejected

D. Allocation Management Module

- Create one allocation for a one-time request if all rules pass
- Create multiple allocations for a recurring request if all occurrences pass
- Update request status to Approved or Rejected

5. Iteration Comparison

Iteration	Request Types	Resource Types	Rules Implemented
Iteration 1	OneTimeRequest	Classroom	AvailabilityRule
Iteration 2	OneTimeRequest	Classroom	AvailabilityRule + CapacityRule
Iteration 3	OneTimeRequest	Classroom + Laboratory	AvailabilityRule + CapacityRule
Iteration 4	OneTimeRequest	Classroom + Laboratory + MeetingRoom	AvailabilityRule + CapacityRule
Iteration 5	OneTimeRequest + RecurringRequest	Classroom + Laboratory + MeetingRoom	AvailabilityRule + CapacityRule
Iteration 6	OneTimeRequest + RecurringRequest	Classroom + Laboratory + MeetingRoom	AvailabilityRule + CapacityRule
Iteration 7	OneTimeRequest + RecurringRequest	Classroom + Laboratory + MeetingRoom	AvailabilityRule + CapacityRule + FeatureRule

6. Updated Class Diagram

Figure 1 shows the updated simplified class diagram for the seventh implementation iteration.

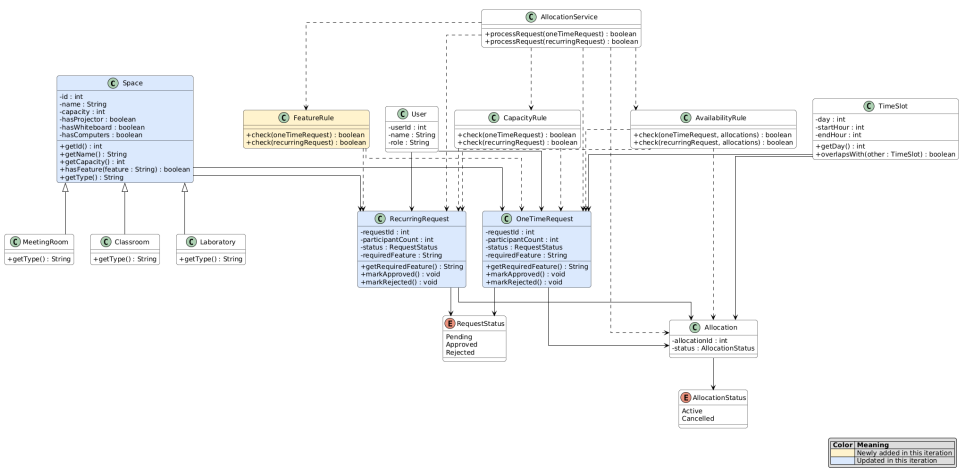

Figure 1: Updated Simplified Class Diagram

7. Updated Workflow Diagram

Figure 2 shows the workflow for the seventh iteration of the space allocation process.

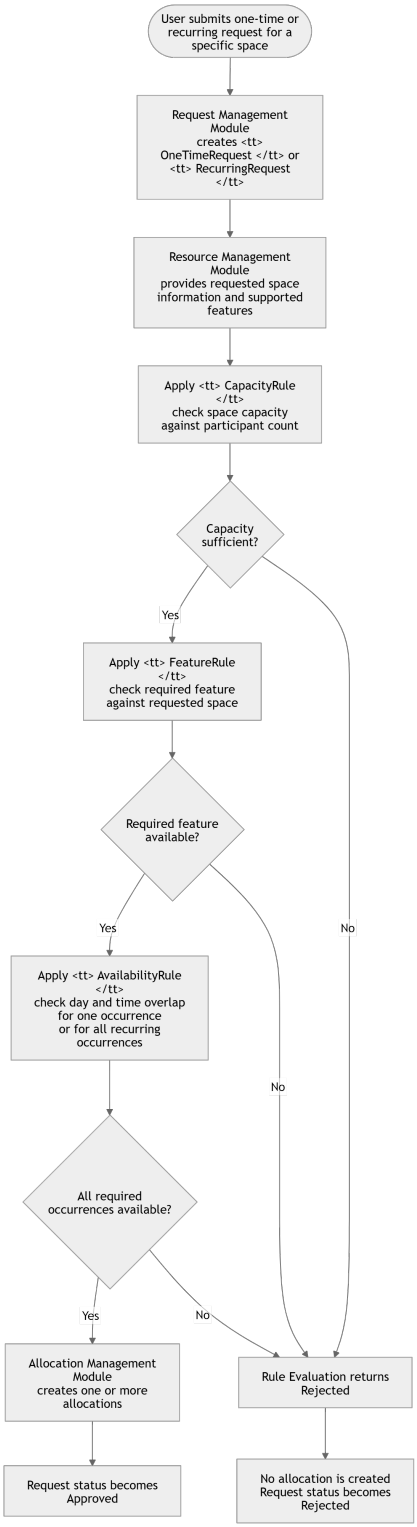

Figure 2: Updated Workflow Diagram

8. Prototype Implementation

To validate the design of this iteration, the C++ prototype was extended by adding supported space features to the `Space` model and a required feature field to both `OneTimeRequest` and `RecurringRequest`. A new decision rule, `FeatureRule`, was introduced to verify that the requested space supports the required feature.

The allocation service continues to support both one-time and recurring requests. It now checks the capacity rule, the feature rule, and the availability rule before creating allocations.

If all rules pass, one or more allocations are created depending on the request type. If one or more rules fail, the request is rejected.

9. Summary

This seventh implementation iteration extends the prototype by introducing required space features while preserving the existing request types, space types, and allocation architecture. The system can now distinguish spaces not only by type, capacity, and time availability, but also by supported features such as projector, whiteboard, and computers.

This iteration demonstrates controlled iterative development by adding a new rule-based constraint without changing the overall structure of the system. Future iterations can continue this approach by extending feature handling or adding richer decision explanations.